COM111 Tutorial 4b

1. For the circuit in Fig. 1, find:
   (a) i(0⁺) and v(0⁺),
   (b) di(0⁺)/dt and dv(0⁺)/dt,
   (c) i(∞) and v(∞).

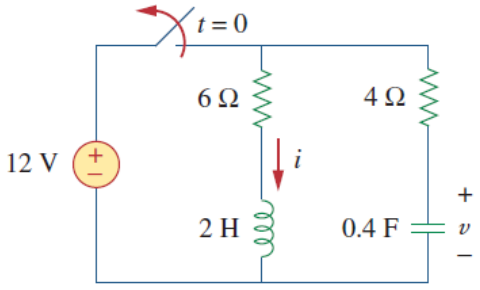


Fig. 1

2. The current in a series *RLC* circuit is described by.

$$\frac{d^2i}{dt^2} + 10\frac{di}{dt} + 25i = 0$$

If $i(0) = 10$ A and $di(0)/dt = 0$, find $i(t)$ for $t > 0$.

3. The differential equation that describes the voltage in an RLC network is

$$\frac{d^2v}{dt^2} + 5\frac{dv}{dt} + 4v = 0$$

Given that $v(0) = 0$, $dv(0)/dt = 10$ V/s, obtain $v(t)$.

4. The switch in Fig. 2 moves from position A to position B at t = 0 (please note that the switch must connect to point B before it breaks the connection at A, a make-before-break switch). Let v(0) = 0, find v(t) for t > 0.

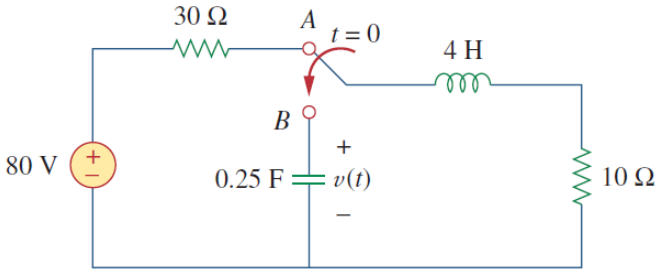


Fig. 2

5. Find $i(t)$ for $t > 0$ in the circuit of Fig. 3.

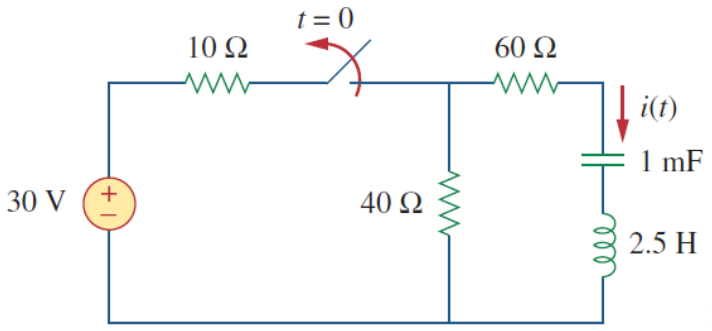


Fig. 3

6. The switch in Fig. 4 moves from position A to position B at t = 0 (please note that the switch must connect to point B before it breaks the connection at A, a make-before-break switch so that the inductor current is not interrupted.). Determine i(t) for t > 0.

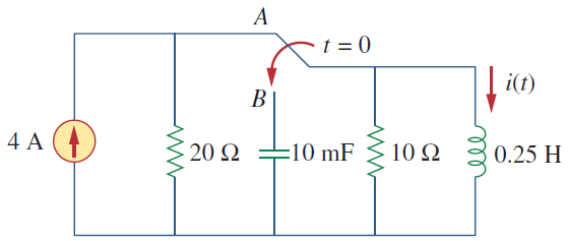


Fig. 4

7. The step responses of a series *RLC* circuit are.

$$v_C = 40 - 10e^{-2000t} - 10e^{-4000t}\ \text{V}, \qquad t > 0$$
$$i_L(t) = 3e^{-2000t} + 6e^{-4000t}\ \text{mA}, \qquad t > 0$$

Find C and determine what type of damping is exhibited by the circuit.

8. For the circuit in Fig. 5, find $v(t)$ for $t > 0$.

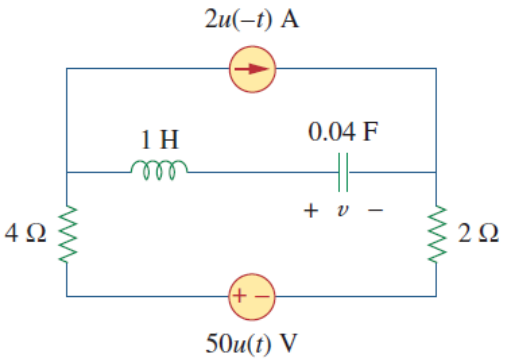


Fig. 5

9. In the circuit of Fig. 6, find v(t) and i(t) for t > 0. Assume v(0) = 0 V and i(0) = 1 A.

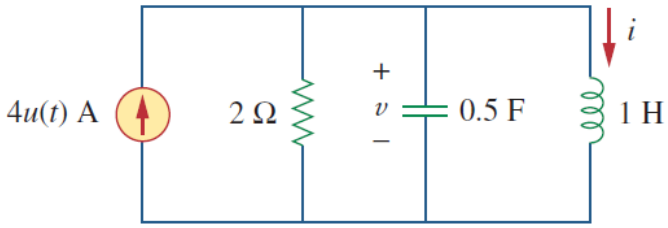


Fig. 6

10. Find the output voltage $v_o(t)$ in the circuit of Fig. 7.

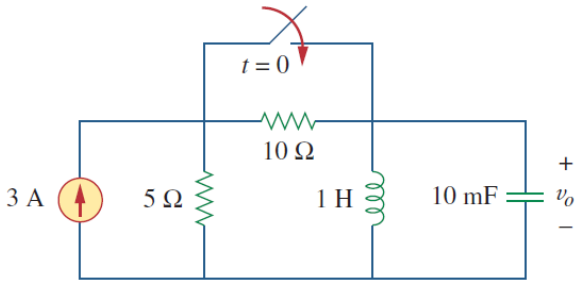


Fig. 7

11. If the switch in Fig. 8 has been closed for a long time before t = 0 but is opened at t = 0, determine:
    (a) the characteristic equation of the circuit,
    (b) i_x and v_R for t > 0.

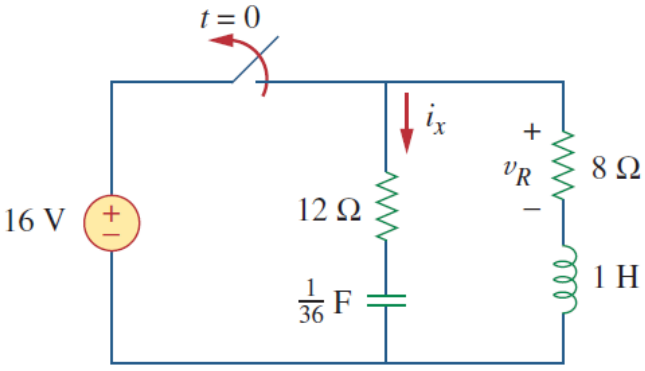


Fig. 8